OpenBattery Hub

OpenBattery Hub is a DIY-friendly IoT platform based on the ESP32-S3, made to work seamlessly with the open-source OpenDTU-OnBattery project. It makes it easy to connect and monitor batteries, solar charge controllers, and inverters—all in one compact device. Whether you're building your own off-grid setup or just love tinkering with energy tech, the Hub is designed to keep things simple, open, and fun to experiment with.

Specifications

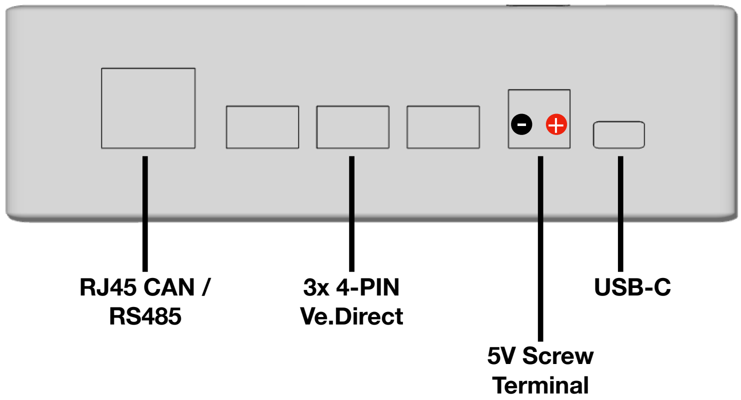
Function	Specification
MCU	ESP32-S3 16MB
CAN Bus / RS-485 Interfaces	SN65HVD230DR / ISL3178EIBZ-T
4-PIN Ve.Direct Interfaces	3x 4 PIN JST PH Interfaces
Power & Programming	USB-C interace, onboard 5V to 3.3V regulator supports max. 1A
Dual Inverter Interface	Nordics NRF24L01+ (HM) / EByte CMT2300a (HMS/T)

What you can do with the OpenBattery Hub

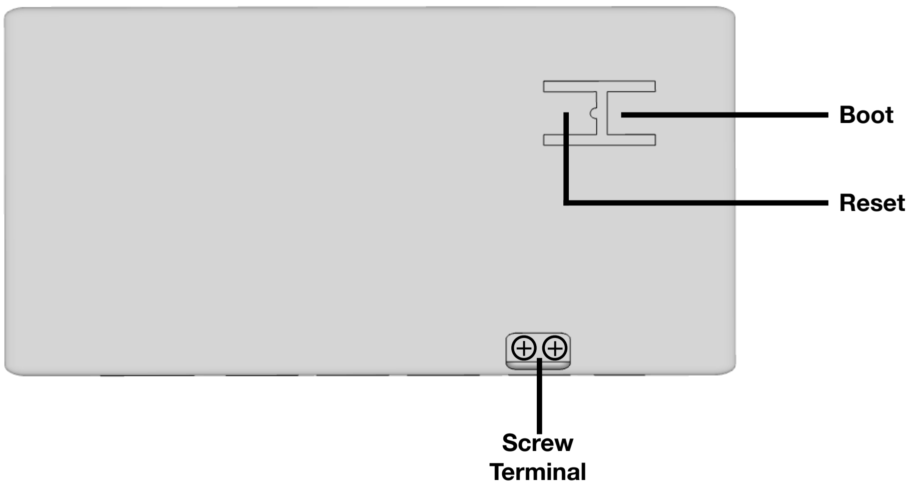
- Unlock the full potential of OpenDTU-OnBattery—all features fully supported
- Connect and control Hoymiles HM, HMS, and HMS-T inverters
- Connect and monitor up to 3 Victron MPPTs
- Interface with popular batteries like Pylontech (CAN-Bus), JK-BMS, and Victron SmartShunt
- Optionally add a compact OLED display for real-time stats right on the device

Ports and Interfaces

Front view



Top view



Getting Started

What you'll need:

- A USB-C power supply (5V / 2A recommended)
- A computer, tablet, or smartphone
- A Wi-Fi connection
- Your Solar System
- Cables according to your Solar System (Ve.Direct, RS-485 Adapter, Cat5 cable etc.)

Power it up
Plug a 5V / 2A USB-C power supply into the DTU's USB-C port.

Connect to the hotspot
Wait a few moments until the DTU creates its own Wi-Fi network called `OpenDTU-xxxxxx`.
Connect to it using the default password: **OpenDTU42**

Open the web interface
Once connected, open a browser and go to **192.168.4.1**

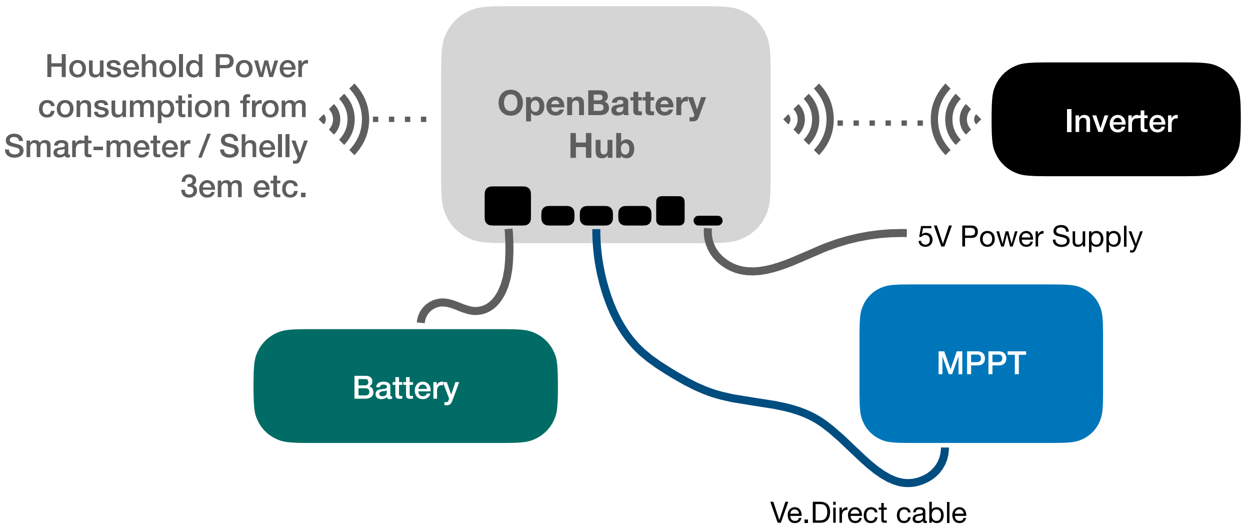
Configure your DTU
Follow the on-screen setup to match your needs.
Be sure to select the correct **configuration profile** for your setup.

Hook up your solar gear
After configuration, connect the DTU to your batteries, inverters, or charge controllers according to the selected profile.

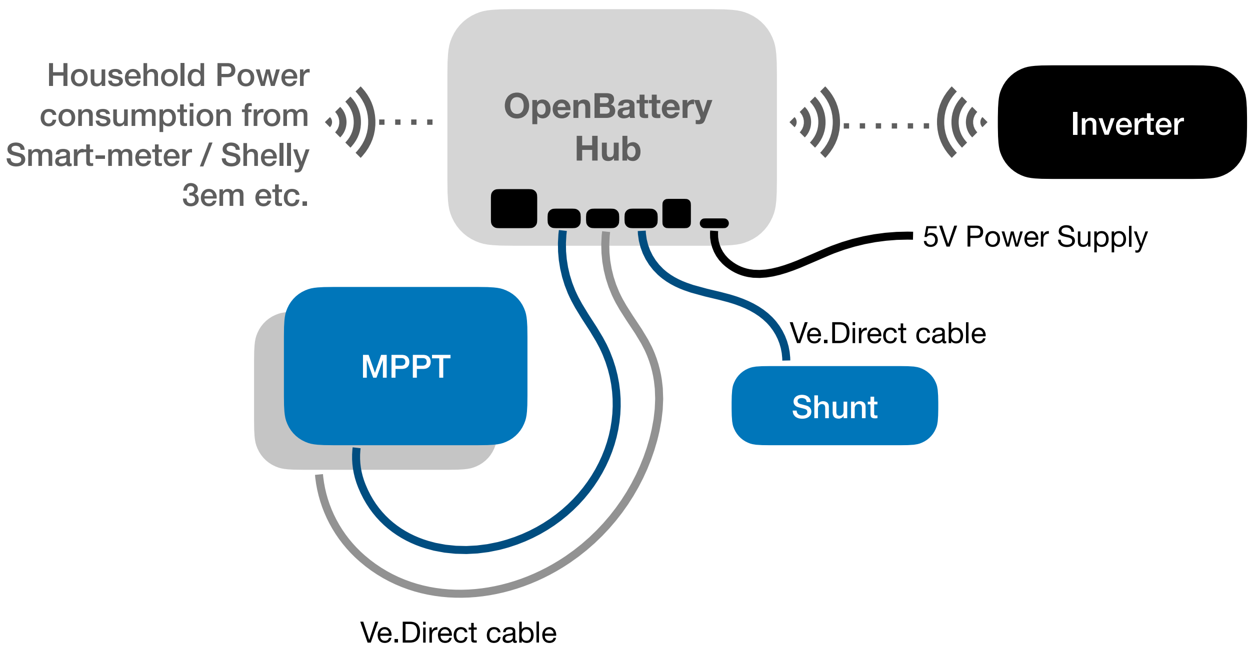
A more comprehensive overview of how to get started with OpenDTU-OnBattery can be found in the internet at https://opendtu-onbattery.net/.

Example Connection Setups

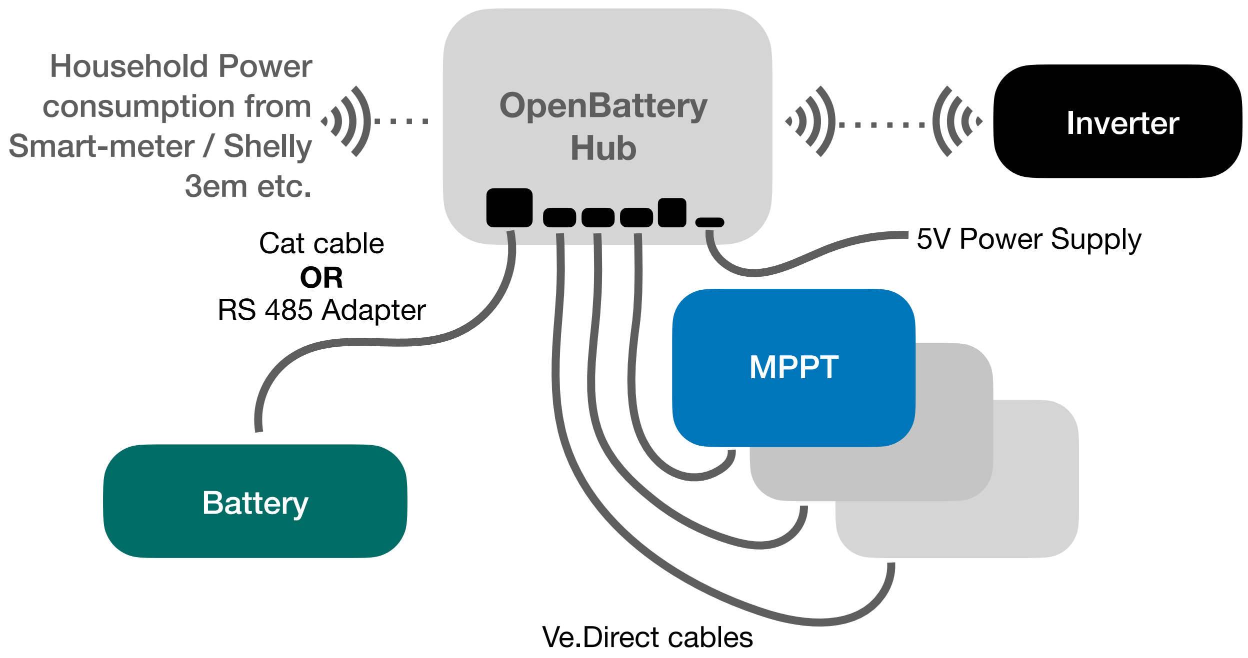
MPPT & CAN / RS-485 Battery



MPPT & SmartShunt Configuration



3x MPPT & CAN or RS-485 Battery



Inverter Support

The OpenBattery Hub is compatible with inverters from Hoymiles, including both HM and HMS/T series. It also works with equivalent models from other manufacturers.
For a full list of supported models and setup instructions, check the OpenDTU documentation.

Solar Charge Controller Support

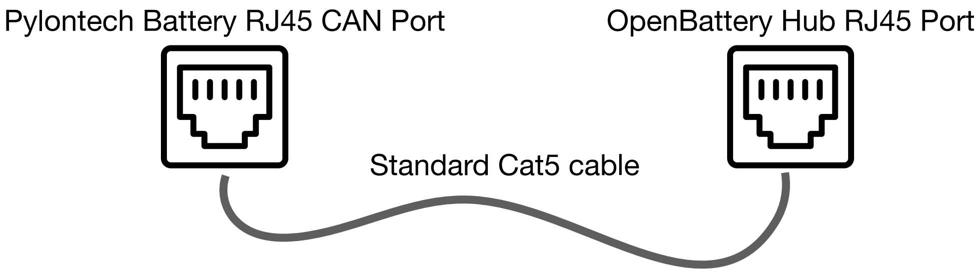
The OpenBattery Hub supports up to three Victron MPPT solar charge controllers via VE.Direct connections using one Victor VE.Direct cable per controller.

- Out of the box, the firmware supports up to 2 MPPTs
- A third MPPT can be added by flashing a special firmware version, available on the OpenDTU-OnBattery GitHub

Battery Interfaces

RJ-45 CAN BUS (Pylontech)

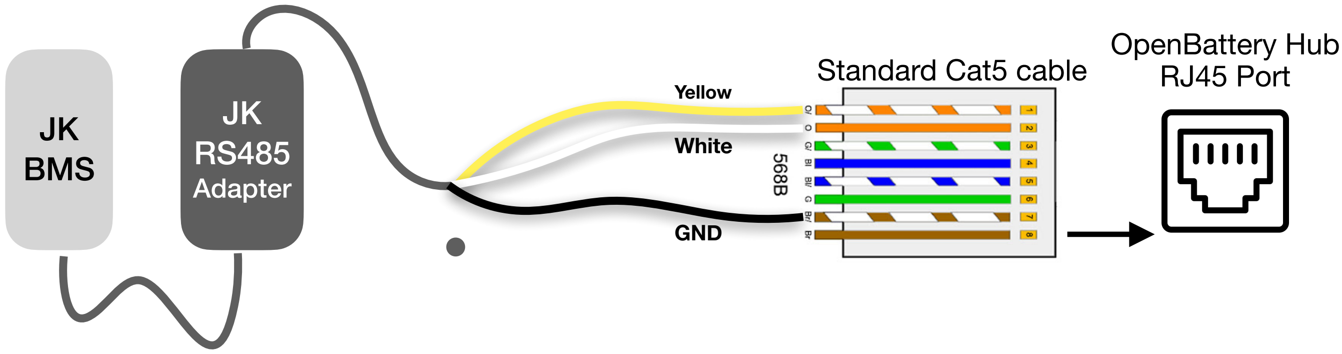
The RJ45 supports CAN Bus interface to Pylontech and compatible Batteries.
A standard RJ45 cable can be used to connect the CAN interface of the battery to the onBattery CAN interface.



Battery Provider: Pylontech via CAN Bus
Profile: 1 or 2 or Z1*

RJ45 RS485 (JK BMS)

The connection between the OpenBattery Hub and a JK BMS managed Battery can be established by using the official JK RS485 Adapter (separate purchase) and a standard Cat5 cable, where the wires 1,2 and 7 will be used.

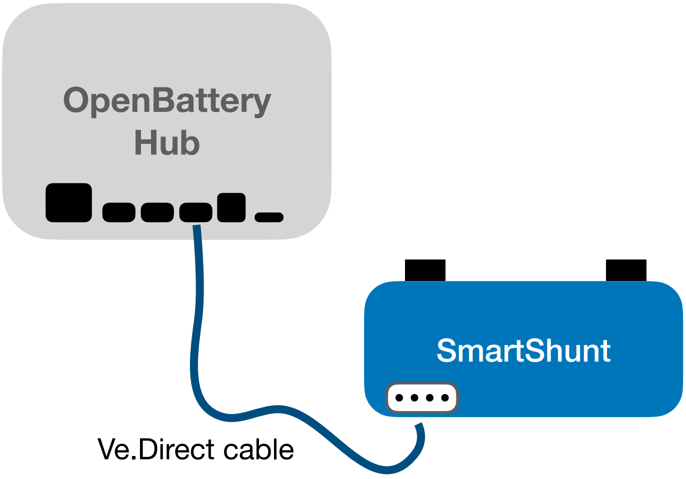


Battery Provider:JiKong (JK) BMS via serial Connection
Profile: 4 or 5 or Z3*

*Requires special firmware

Ve.Direct SmartShunt

TO connect the OpenBattery Hub to a SmartShunt managed Battery a standard Victron Ve.Direct Cable is being used (separate purchase). For configuration of the SmartShunt please refer to the official manual-



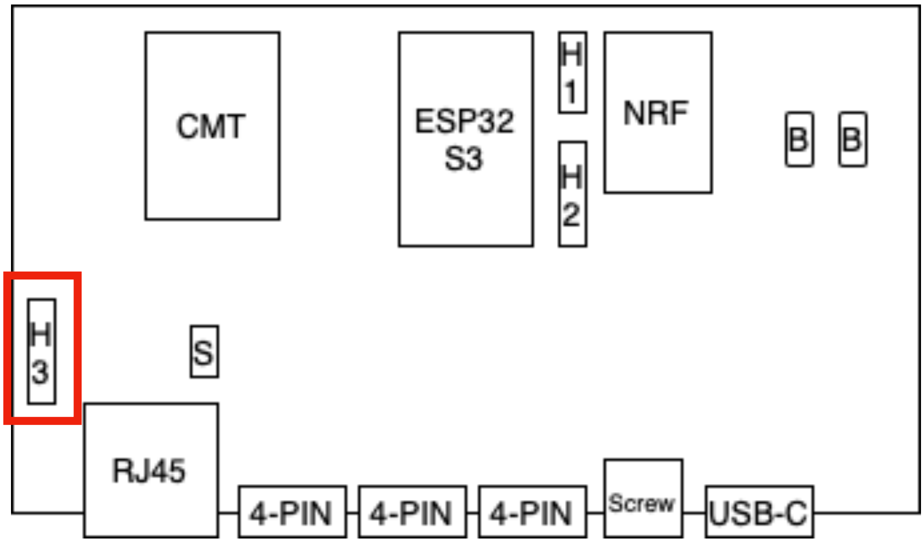
Battery Provider: Victron SmartShunt per Ve.Direct Interface
Configuration Profile: Depending on the number of MPPTs, profile 3 or Z2*

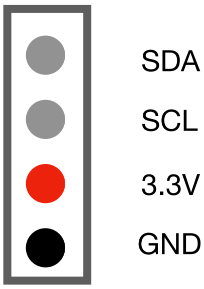
*Requires special firmware

Optional PIN Connectors

To use the optional PIN Connectors the DTU has to be opened. To do so unscrew the 4 M3 screws on the bottom of the the DTU.

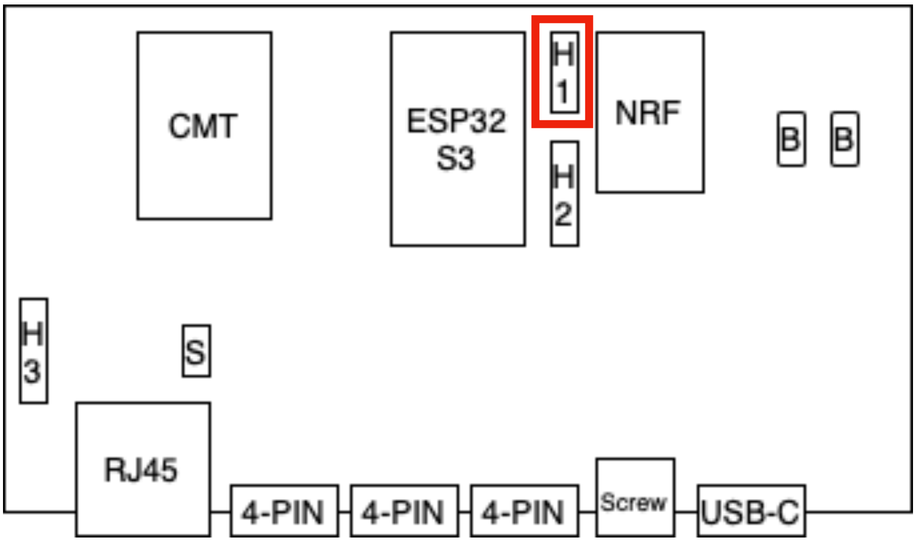
OLED Display





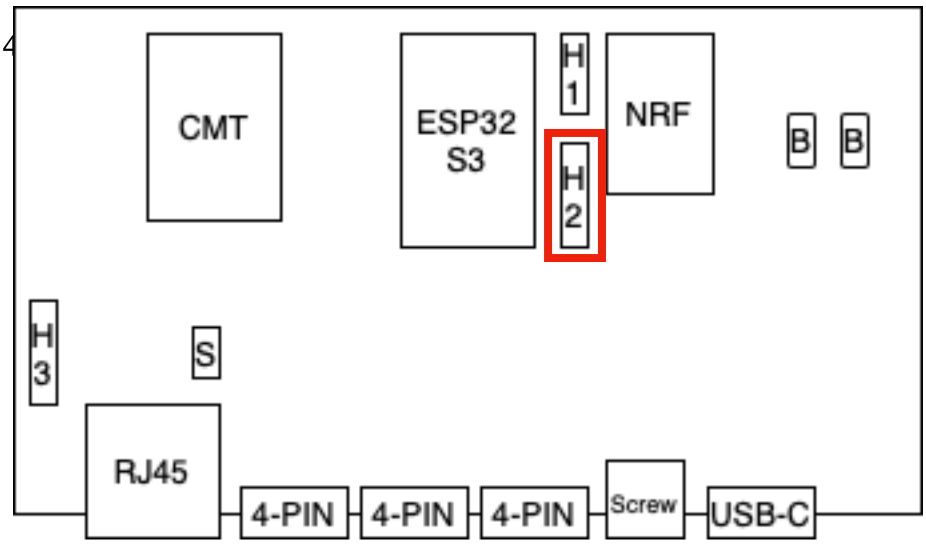
Serial Header

Can be used to connect to the Serial console of OpenDTU-OnBattery.



Unmapped PINs

GPIO 39, 40, 4



LEDs

Antenna

CAN Bus Termination

Power Supply

Buttons